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Chapter

PSYCH 301CL
Cognitive Processes

Lecture 15:
Judgment, Decision Making, and Reasoning

Fall 2026

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Chapter

Decisions and Reasoning

- Decisions: The process of making choices between alternatives
- Reasoning: The process of drawing conclusions

- Decisions are based on judgments we make, and applying these judgments can involve various reasoning processes

- One of the primary mechanism involved in making judgments in inductive reasoning
  - Reasoning based on observations
  - Reaching conclusions from evidence

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Inductive Reasoning

- Inductive reasoning is the basis of scientific investigations
  - In science: observations are made, data are collected, and conclusions are drawn

– In inductive reasoning conclusions we reach are *probably*, but not *definitely*, true
  - Conclusions reached are suggested with various degrees of certainty

– Strength of argument
  - *Representativeness* of observations
  - *Number* of observations
  - *Quality* of observations



Sometimes one very compelling high quality observation can carry a lot of weight.

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Inductive Reasoning

- Used to make scientific discoveries
  - Hypotheses and general conclusions

- Used in everyday life
  - Make a prediction about what *will happen* based on observation about what *has happened* in the past
  - We make predictions and choices based on past experience, especially when predictions are based on familiar situations
    - Provides a mechanism for using past experience to guide present behavior
  - Some types of inductive reasoning happen automatically

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Heuristics

- When people use past experience to guide present behavior, they often use shortcuts to help them reach conclusions rapidly

- “Rules of thumb” that are likely to provide the correct answer to a problem, but are not foolproof

- Two more commonly used heuristics include the *availability* heuristic and the *representativeness* heuristic

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Heuristics

- Availability heuristic: events more easily remembered are judged as being more probable than those less easily remembered

- Problem lies in factors that make events stand out in memory (e.g., distinctiveness, emotional salience, frequency of encoding / retrieval)

- Availability is used to judge the probability of events
  - You use availability by trying to call to mind examples
  - The more examples you think of, the more probable you judge the event to be

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Heuristics

- Which are more prevalent in English, words that being with the letter *r* or words in which *r* is the third letter?

- 70% respond that more words begin with *r*, even though in reality three times more words have *r* in the third position

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Heuristics

- Some possible causes of death are listed below in pairs. Within each pair, which cause of death do you consider to be more likely for people in the United States? That is, if you randomly picked someone in the US, would that person be more likely to die next year from cause A or cause B?

MORE LIKELY	LESS LIKELY	PERCENT PICKING LESS LIKELY
Homicide (20)	Appendicitis	9
Drowning (5)	Auto–train collision	34
Asthma (920)	Botulism	41
Asthma (20)	Tornado	58
Appendicitis (2)	Pregnancy	83

- This is due to availability of the causes.
  - E.g. when people die of botulism, it is front-pages news
  - Deaths from asthma go virtually unnoticed

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Heuristics

- Representativeness heuristic: people often make judgments based on how much one event resembles another event
  - The probability that A is a member of class B can be determined by how well the properties of A resembles properties normally associated with class B

- America has about 2.2M farms and 120k libraries
- We randomly pick one male from the population of the US: That male, Robert, wears glasses, speaks quietly, and reads a lot. Is it more likely that Robert is a librarian or a farmer?

- More S predict that Robert is a librarian
- ➔ they ignore the base rate of farmers and librarians in the US

Heuristics

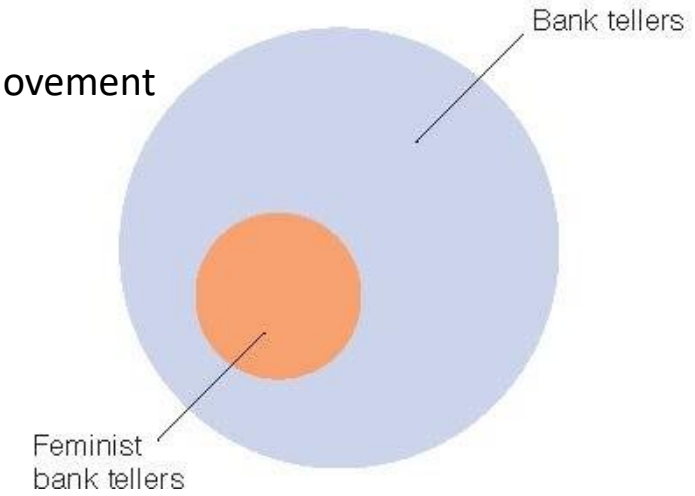
- Rule S follow:
  - Use base rate information if it is all that is available
  - Use descriptive information if available and disregard base rate information
- In a group of 100 people, there are 70 lawyers and 30 engineers. What is the chance that if we pick one person from the group at random that the person will be an engineer?
- ➔ S get this right
- Jack is a 45-year-old man. He is married and has four children. He is generally conservative, careful, and ambitious. He shows no interest in political and social issues and spends most of his free time on his many hobbies, which include home carpentry, sailing, and mathematical puzzles.
- ➔ S say Jack is an Engineer

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Heuristics

- Conjunction rule: probability of two events cannot be higher than the probability of the single constituents

  - Linda is 31 years old, single, outspoken, and very bright. She majored in philosophy. As a student, she was deeply concerned with issues of discrimination and social justice, and also participated in antinuclear demonstrations. Which of the following alternatives is more probable?
    1. Linda is a bank teller
    2. Linda is a bank teller and is active in the feminist movement
  - 82% pick 2



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Heuristics

- People also make errors in reasoning by ignoring the importance of the size of the sample on which observations are based

  - A certain town is served by two hospitals. In the larger hospital about 45 babies are born each day, and in the smaller hospital about 15 babies are born each day. As you know, about 50% of all babies are boys. However, the exact percentage varies from day to day. Sometimes it may be higher than 50%, sometimes lower. For a period of 1 year, each hospital recorded the days on which more than 60% of the babies born were boys. Which hospital do you think recorded more such days?
    1. The larger hospital
    2. The smaller hospital
    3. About the same?

  - S: 1 - 22% , 2 - 22%, 3 - 56%

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Heuristics

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    3. About the same?

  - S: 1 - 22% , 2 - 22%, 3 - 56%
  - ➔ correct answer is 2

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Heuristics

- Law of large numbers: the larger the number of individuals randomly drawn from a population, the more representative the resulting group will be of the entire population
  - People often assume that representativeness holds for small samples, and this results in errors in reasoning

Heuristics

- Illusory correlations: correlation appears to exist, but either does not exist or is much weaker than assumed

- They can occur when we expect two things to be related, so we fool ourselves into thinking they are related even when they are not
  - Stereotypes: Oversimplified generalizations about a group or class of people that often focuses on the negative
    - A stereotype about a group may lead people to pay particular attention to behaviors associated with that stereotype
    - This attention creates an illusory correlation that reinforces the stereotype
    - Related to the availability heuristic because selective attention to the stereotypical behaviors makes these behaviors more “available”

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Confirmation Bias

Lord and coworkers (1979)

- Had those in favor of capital punishment and those against it read the same article
- Those in favor found the article in favor
- Those against found the article against

The “myside” bias: tendency for people to generate and evaluate evidence and test their hypotheses in a way that is biased toward their own opinions and attitudes

➔ S focused on information that agreed with their beliefs and disregarded information that didn’t

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Confirmation Bias

- The myside bias is a type of confirmation bias

- Confirmation bias: tendency to selectively look for information that conforms to our hypothesis and overlook information that argues against it
  - Yo… Yo…th yo… tell yo…dent that you have discovered the rule, you are to write it down and tell me what it is.

Confirmation bias acts like a pair of blinders: we see the world according to rules we think are correct and are never dissuaded from this view because we seek out only evidence that confirms our rule

  - 2, 4, 6
  - Then S started generating their own sets of three numbers
  - For each set they received feedback, but only to whether their numbers fit his rule or not
  - The most common hypothesis: increasing intervals of two
  - Actual rule: three numbers in increasing order of magnitude

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Decision-Making

- Decision making: making judgments that involve choices between different courses of action

- Utility: outcomes that are desirable because they are in the person's best interest

- Expected utility theory:
  - People are *rational*
  - If they have all relevant information, they will make a decision that results in the maximum expected utility
  - Maximum monetary payoff

Decision-Making

- Advantages for utility approach
  - Specific procedures to determine the "best choice"

- Problems for utility approach
  - Not necessarily money, people find value in other things
  - Many decisions do not maximize the probability of the best outcome
    - ➔ people are not utility maximization machines
    - e.g. people gamble in casinos all the time, even though they know that the odds are very much against them

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Decision-Making

- S were given the opportunity to earn up to $7, by receiving $1 every time they drew a red jellybean
- Each time they were given a choice between two different bowls
- Which one do you choose?

Most S chose 7/100

➔ Their choice was influenced by considerations other than probabilities



odds are 1/10



odds are 7/100 (or 0.7/10)

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Decision-Making

- After 9/11 a decrease in air travel and increase in driving occurred
  - Even though the prob of dying in a car accident if far greater than in flying
  - The number of Americans who lost their lives on the road to avoid flying is higher than the total number of passengers killed on the four hijacked flights

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An Example Dilemma

- A convoy of food trucks is on its way to a refugee camp during a famine. (Airplanes cannot be used.) You find that a second camp has even more refugees. If you tell the convoy to go to the second camp instead of the first, you will save 1000 people from death, but 100 people in the first camp will die as a result.

  - Would you send the convoy to the second camp?
    - U(notReRoute)=100-1000
    - U(Reroute)=1000-100
  - Ritov and Baron (1999): **63%** did not choose to send the convoy to the second camp
    - **12%** responded that they would not change the route of the convoy even if only 1 person in the first camp would die

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Tetlock, Kristel, Elson, Green, Lerner (2000)

Sacred Value Protection Model

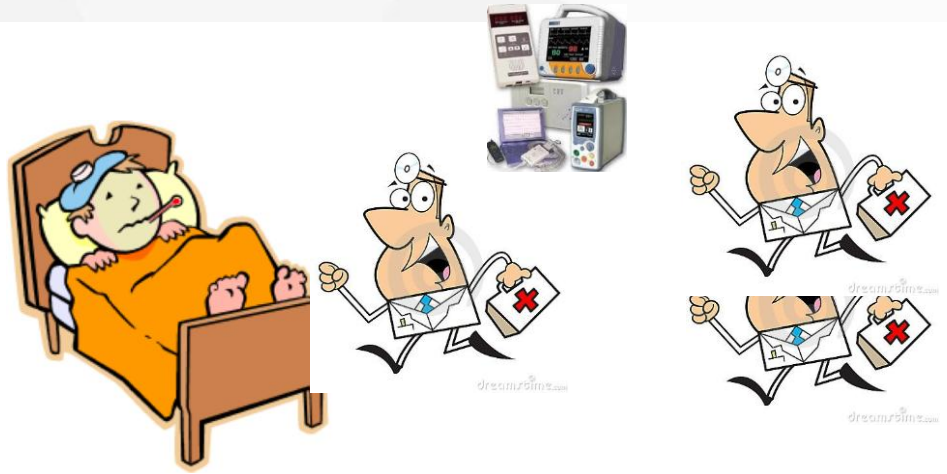
- People want to protect what they hold sacred
  - Sacred values are compartmentalized separately from secular values.
  - When the two domains are intermingled, people feel threatened and they respond with moral outrage (Taboo).
  - In contrast, when two sacred values are compared with each other, people are less morally outraged (Tragic).

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Tetlock, Kristel, Elson, Green, Lerner (2000)

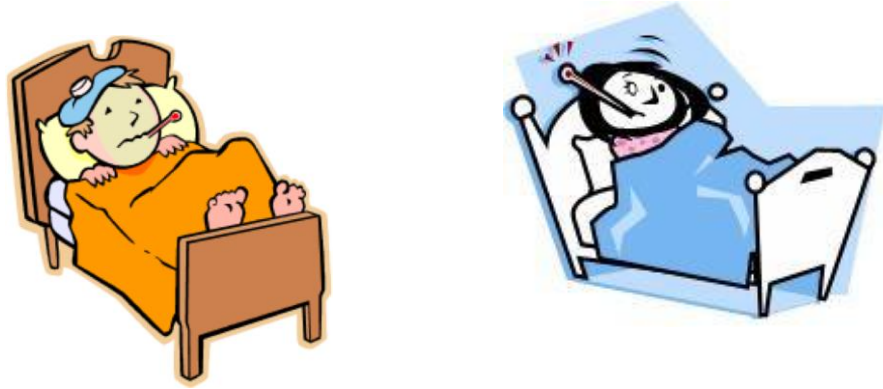
- Taboo Tradeoff Scenario

Robert can save the life of Johnny, a five year old who needs a liver transplant, but the transplant procedure will cost the hospital $1,000,000 that could be spent in other ways, such as purchasing better equipment and enhancing salaries to recruit talented doctors to the hospital. Johnny is very ill and has been on the waiting list for a transplant but because of the shortage of local organ donors, obtaining a liver will be expensive. Robert could save Johnny's life, or he could use the $1,000,000 for other hospital needs.

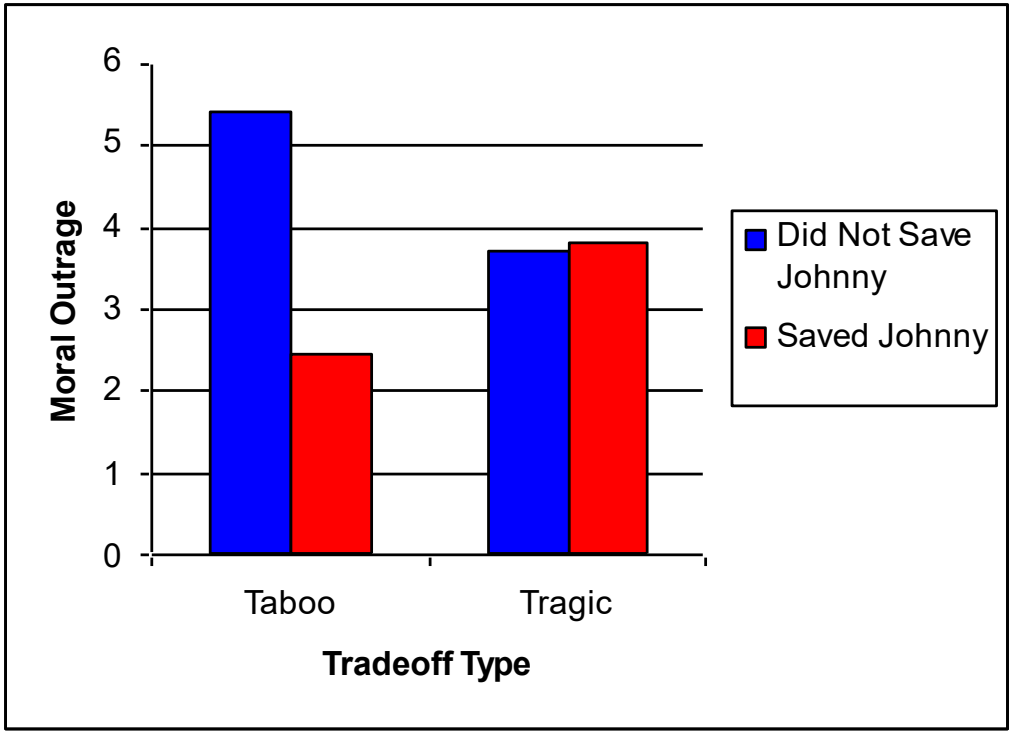


- Tragic Trade-off Scenario

There are two children who need a liver transplant, but only one liver.



Tetlock, Kristel, Elson, Green, Lerner (2000)



Ginges, Atran, Medin & Shikaki (2007)

- Extended previous findings to a more realistic setting.
  - Interviewed Israeli settlers (n=601), Palestinian refugees (n=535) and Palestinian students who were affiliated with various political groups (719)
- Offered three types of tradeoff deals as possible solutions to the conflict.

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Ginges, Atran, Medin & Shikaki (2007)

Taboo: Palestinians would recognize the sacred and historic right of the Jewish people to Israel. There would be two states: a Jewish state of Israel and a Palestinian state in 99% of the West Bank and Gaza.

Taboo+: On their part, Israel will pay Palestine 1 billion U.S. dollars a year for 10 years.

Symbolic: On their part, Israel will recognize the historic and legitimate right of the Palestinians to their own state and would apologize for all of the wrongs done to the Palestinian people.

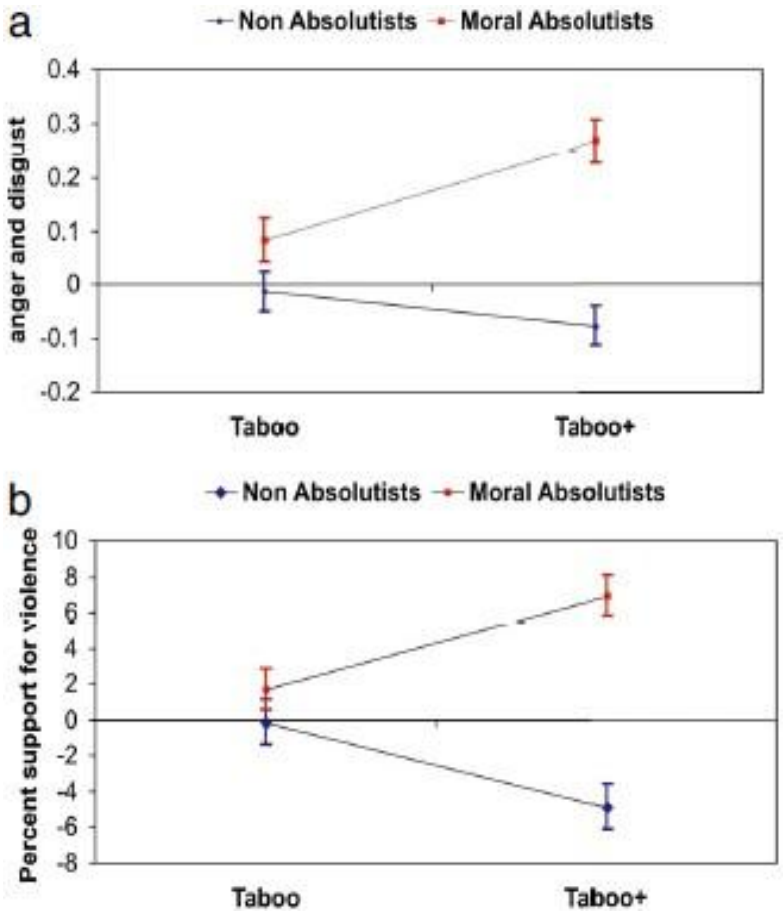
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SV Measure

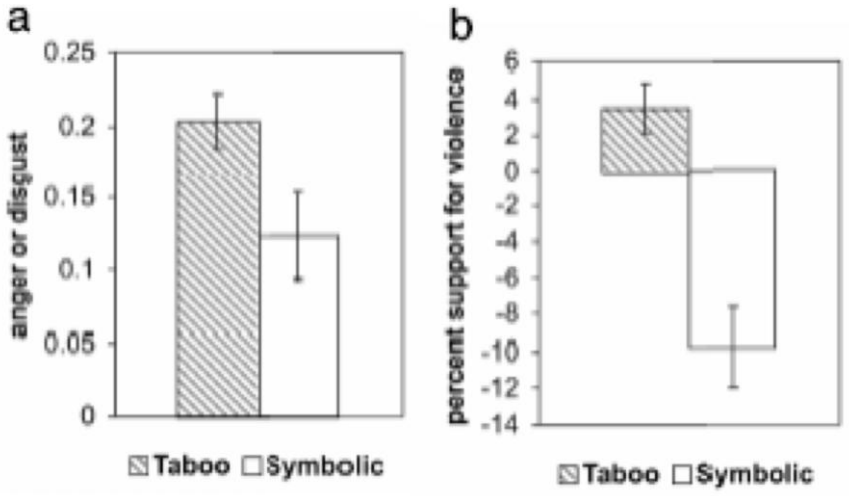
- moral-absolutists: transformed the issues under dispute into sacred values
  - Among settlers: 46% of the sample
    - believed that it was never permissible for the Jewish people to ‘‘give up’’ part of the ‘‘Land of Israel’’ no matter how great the benefit.
  - Among Palestinian students 54%
    - treated both the principle of the ‘‘right of return’’ and Jerusalem as sacred values.

- non-absolutists: had strong preferences against compromise over these issues but did not regard them as sacred values

Ginges, Atran, Medin & Shikaki (2007)



Taboo vs. Taboo +



Taboo vs. Symbolic (Tragic)

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Sacred Values in Intergroup Conflicts

Investigate the impact of monetary incentives/disincentives in negotiations involving sacred values.

In two separate studies (Dehghani, Atran et al. 2010; Dehghani, Iliev et al. 2009):

- How material incentives and disincentives, such as economic sanctions, psychologically affect the targeted population and potentially influence behavior

- Investigated whether Iranians who treat the Iranian nuclear program as a sacred value would be amenable to material incentives/disincentives to compromise







The New York Times











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Experiments

- 1997 Iranian participants in an online study conducted in Farsi
- Dependent variables: people's approval, people's anger, misleading factor

- The sacred value (SV) measure (based on Baron and Spranca's 1997 measure):
  What do you think about Iran giving up its nuclear energy/weapons?
  - a. I think this definitely needs to happen.
  - b. I do not object to this.
  - c. This should be done only if the benefits of stopping the program are great enough.
  - d. This shouldn't be done no matter how great the consequences

- Participants who answered 'd' were considered to have a SV (14% for nuclear energy, 13% nuclear weapons)

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Scenarios

- 2 (Nuclear program: nuclear energy vs nuclear weapons) * 3 (Type of deal: Taboo, Taboo+, Taboo-) design

- Scenarios:
  - Taboo: Iran will give up its program for developing nuclear energy(or weapons) and surrender the current nuclear facilities to the UN; in return Israel will give up their program for developing nuclear energy (or weapons) and surrender the existing nuclear facilities to the UN.
  - Taboo+: taboo + In return, United Nations will give $40 billion dollars in economic funds to Iran.
  - Taboo-: Taboo + You know that if Iran does not accept this deal, United Nations will impose additional sanctions on Iran.

- Participants were asked:
  - Do you personally approve of this deal?
  - If the Iranian government approves such a deal do you think that people will be angry?
  - If such deal is really proposed to Iran do you find it misleading?

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Results:

